Calculation Supplement — Document 1

Exhaustive Descriptive Statistics

Updated version: nine navigation methods

343 valid runs / 9 methods / 4 scenarios / 15 metrics

Step-by-step statistical analysis

of social navigation in an agricultural digital twin

Purpose. To provide reproducible support for the descriptive statistics of the consolidated Unity–MATLAB corpus. This revision expands the original five-method, 184-successful-run document to nine methods and 343 valid records by adding baselines B1–B4. Every continuous statistic is explicitly success-conditional; failures remain represented in the success rate.

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Contents

1 Descriptive-statistics framework

Let $X = \{x_1, x_2, \ldots, x_n\}$ be a sample of a scalar metric from the successful runs of one method. The arithmetic mean, sample variance, and sample standard deviation are

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \tag{1}$$

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2, \qquad s = \sqrt{s^2}. \tag{2}$$

Bessel's correction makes s^2 unbiased for the population variance under independent sampling. The coefficient of variation and interquartile range are

$$CV = \frac{s}{\bar{x}}, \qquad IQR = Q_{75} - Q_{25}. \tag{3}$$

Percentiles use linear interpolation. When $\bar{x} = 0$, CV is undefined and must not be interpreted.

1.1 Success conditioning

A run succeeds when it completes $A \to B \to A$ before $T_{\max} = 180$ s. Continuous metrics are summarized only over successful runs, preventing a timeout from being treated as a completed trajectory. All valid records, successful or otherwise, contribute to $\eta_s = N_{success}/N_{valid}$.

2 Updated experimental design

Five in-house variants (M0–M4) and four external baselines were evaluated: Social DWA (B1), ORCA/RVO (B2), Social Force Model (B3), and CBF-Social-DWA (B4). The original study contributed 223 runs for M0–M4/B1; this extension adds 120 runs for B2/B3/B4 (40 each), for a corpus of 343 runs across nine methods. One B1 run was excluded for data integrity ($n = 39$ valid). Binary supervisors M1/M2 fail 8/10 E4 runs by deadlock; B4 fails 2/10 in E2 and 1/10 in E4 because of braking-loop timeouts. Total corpus stated by the paper: 343 runs.

Table 1: Executed matrix: successful/planned runs.

Scn.	M0	M1	M2	M3	M4	B1	B2	B3	B4
E1	10/10	10/10	10/10	10/10	10/10	9/10	10/10	10/10	10/10
E2	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10	8/10
E3	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10	10/10
E4	10/10	2/10	2/10	10/10	10/10	7/9	10/10	10/10	9/10
Total	40	32	32	40	40	36/39	40	40	37/40

3 Recorded metrics

The 15 scalar metrics are minimum, mean, and fifth-percentile distance; personal- and intimate-zone time; mission time; path length and path efficiency; mean, maximum, and standard deviation of speed; maximum numerical pseudo-acceleration and RMS acceleration; stopped time; and number of stops. Success rate is handled separately as a binary outcome.

4 Global descriptive statistics

Tables 2 and 3 report mean ± sample standard deviation, conditional on success.

Table 2: Safety and zone-occupancy metrics (mean ± SD, success-conditional). M0–M4/B1 are from the original study; B2/B3/B4 were computed from the 120 new per-run records.

Method	$d^\star$ [m]	$\bar{d}$ [m]	$T_{personal}$ [s]	$T_{intimate}$ [s]	$T_{mission}$ [s]	N_{stop}
M0 ($n = 40$)	0.563 ± 0.115	4.98 ± 3.09	25.4 ± 17.0	0.02 ± 0.09	93.6 ± 2.5	0.2 ± 0.4
M1 ($n = 32$)	0.750 ± 0.097	5.95 ± 2.80	20.3 ± 14.0	0.00 ± 0.00	107.9 ± 21.0	9.9 ± 15.3
M2 ($n = 32$)	0.745 ± 0.111	6.93 ± 2.15	12.4 ± 5.4	0.00 ± 0.00	109.3 ± 24.2	3.2 ± 1.6
M3 ($n = 40$)	1.004 ± 0.209	6.18 ± 2.31	7.6 ± 6.7	0.00 ± 0.00	115.0 ± 33.3	0.0 ± 0.2
M4 ($n = 40$)	0.846 ± 0.189	5.72 ± 2.72	11.7 ± 11.4	0.00 ± 0.00	110.6 ± 32.1	0.2 ± 0.4
B1 ($n = 36$)	0.891 ± 0.495	7.45 ± 4.62	3.0 ± 2.1	0.16 ± 0.50	124.6 ± 18.1	2.1 ± 1.0
B2 ($n = 40$)	0.249 ± 0.262	4.62 ± 3.88	27.5 ± 15.8	1.13 ± 0.82	97.8 ± 2.4	0.08 ± 0.27
B3 ($n = 40$)	0.730 ± 0.288	6.38 ± 2.61	4.99 ± 3.32	0.16 ± 0.37	106.2 ± 4.9	0.18 ± 0.45
B4 ($n = 37$)	1.280 ± 0.292	7.50 ± 2.29	0.45 ± 0.78	0.02 ± 0.12	113.4 ± 9.8	1.35 ± 1.64

Table 3: Efficiency and smoothness metrics (mean ± SD, success-conditional). a_{max}^{num} carries the cross-source caveat stated in the paper; σ_v is the primary smoothness metric.

Method	L [m]	$\bar{v}$ [m/s]	σ_v [m/s]	v_{max} [m/s]	a_{max}^{num} [m/s^2]	T_{stop} [s]
M0	93.26 ± 2.54	0.998 ± 0.002	0.066 ± 0.014	1.88 ± 0.56	12.43 ± 7.01	0.04 ± 0.09
M1	81.22 ± 24.12	0.755 ± 0.173	0.378 ± 0.067	1.79 ± 0.55	11.28 ± 6.24	22.67 ± 18.93
M2	82.06 ± 21.20	0.754 ± 0.117	0.410 ± 0.057	1.83 ± 0.49	11.01 ± 5.59	26.00 ± 14.47
M3	89.12 ± 21.37	0.790 ± 0.112	0.243 ± 0.026	1.52 ± 0.51	9.37 ± 4.99	0.01 ± 0.05
M4	85.98 ± 19.95	0.796 ± 0.121	0.195 ± 0.029	1.61 ± 0.39	10.28 ± 4.15	0.04 ± 0.10
B1	98.45 ± 4.58	0.800 ± 0.092	0.238 ± 0.055	1.48 ± 0.42	1.32 ± 0.78†	2.62 ± 2.98
B2	95.36 ± 1.19	0.974 ± 0.016	0.092 ± 0.022	1.61 ± 0.56	10.49 ± 6.67	0.02 ± 0.06
B3	97.63 ± 1.40	0.918 ± 0.034	0.163 ± 0.044	1.00 ± 0.00	3.91 ± 1.25	0.19 ± 0.62
B4	102.08 ± 4.90	0.906 ± 0.052	0.228 ± 0.040	1.77 ± 0.59	9.86 ± 6.77	1.15 ± 2.67

5 Scenario-resolved safety

The pooled ORCA/RVO mean hides critical heterogeneity: B2 enters the intimate zone ($d_I = 0.45$ m) in E1 and E2, whereas B4 maintains an approximately uniform separation.

Table 4: Mean minimum distance by scenario for the new baselines.

Method	E1	E2	E3	E4	Pooled
B2 (ORCA)	**0.065**	**0.059**	**0.248**	0.623	0.249
B3 (SFM)	0.551	0.740	0.933	0.697	0.730
B4 (CBF)	1.298	1.268	1.334	1.211	1.280

6 Worked calculation

For five illustrative B3–E1 values of $d^\star$, $X = \{0.48, 0.52, 0.55, 0.58, 0.62\}$ m:

$$\sum x_i = 2.75, \qquad \bar{x} = 2.75/5 = 0.550 \text{ m}, \tag{4}$$

$$SS = \sum (x_i - \bar{x})^2 = 0.0116, \qquad s^2 = 0.0116/4 = 0.0029 \text{ m}^2, \tag{5}$$

$$s = 0.0539 \text{ m}, \qquad CV = s/\bar{x} = 9.8\%. \tag{6}$$

The consolidated B3–E1 mean is 0.551 m. The example reproduces its scale and demonstrates the procedure; it does not replace the computation over all ten runs.

7 Integrity and reproducibility

All 120 new runs passed the integrity checks stated by the paper: `PacketsInvalid=0`, no NaNs, and standard decimal locale. One B1 run was excluded for data integrity. Per-run scalars are available for B2/B3/B4; published aggregates were reused for M0–M4/B1. The cross-source comparability caveat for a_{max}^{num} remains explicit.

8 Conclusions

1. The corpus stated by the paper contains 343 runs across nine methods; M1/M2 exhibit operational deadlock in E4.
2. B4 has the largest pooled minimum separation (1.280 m) but loses operational continuity in three runs.
3. B2 is fast and smooth but enters the intimate zone in E1/E2; pooled means alone are insufficient for safety assessment.
4. Within M0–M4, M4 combines zero intimate-zone time, low speed variability, and shorter paths than M3, whereas M3 retains the larger mean minimum separation.
5. These results are descriptive and success-conditional; statistical significance and effect sizes belong to the subsequent inferential documents.